

VISHAL DHANASEKARAN

✉ 2005vishaldhanasekaran@gmail.com ☎ +91 8220661814 📧 Vishal D 🌐 VishalDhanasekaran

EDUCATION

GEETHANJALEE MATRIC HR SECONDARY SCHOOL 📄, HSC | Percentage: 85.3% May 2022

COIMBATORE INSTITUTE OF TECHNOLOGY 📄,
M.Sc., Software Systems (Integrated) | CGPA: 8.18 (till 6th Semester) October 2022 – Present

PROFESSIONAL EXPERIENCE

Software Development Engineer 📄, Asthra Medtech Pvt Lmt June 2025 –
November 2025
Coimbatore, On-Site

Ultrasound Scan Report Generating Software

Tech Stack: Next.js, Express.js, Postman

- Developed a **web-based platform** that allows doctors to upload **DICOM medical scans** and automatically generate structured reports using **OCR**.
- Implemented a **report editor** with MPIN-secured digital signatures for secure publishing.
- Reduced report preparation time from **60 minutes to 5 minutes**, significantly boosting doctors' productivity.

Project Management System

Tech Stack: Next.js, Express.js, MySQL, Docker, AWS RDS, Bruno

- Designed and developed **RESTful APIs** with **JWT-based authentication** for secure, role-based routing.
- Built features for **live task tracking, schedule variance monitoring, and resource allocation**, improving efficiency for both developers and project managers.

PROJECTS

Melinia'26 📄, | bun | hono | vite + react (typescript) January 2025 –
February 2026

- Architected and deployed a full-stack event management platform using **Bun, Hono, React, Vite, and TypeScript** in a **monorepo architecture**.
- Successfully handled 2000+ users and event registrations during the tech fest.
- Integrated **Redis + BullMQ for asynchronous job processing**, and designed **PostgreSQL** database schemas to support **scalable** event registrations, payment tracking, and transactional email workflows.
- Implemented **production-grade infrastructure** with **Docker, Caddy reverse proxy**, GitHub Actions CI/CD, and **Prometheus + Grafana monitoring**, deploying the entire system on a **self-managed VPS** with a **self-hosted Mailcow SMTP server** for reliable email delivery and system observability.

Chotta Chat 📄, | Next.js | Ollama | Vercel's AI SDK August 2025

- A lightweight chat application that enables seamless interaction with locally hosted AI models via **Ollama**.
- Features a clean, responsive UI with **real-time streaming responses** for an intuitive user experience.

FOSStream 📄, | React.js July 2025

- A single-page application (SPA) that aggregates **FOSS news** from multiple blogging platforms.
- Implements **infinite scrolling** with smooth, efficient data fetching and a responsive interface for uninterrupted browsing.

DataDashBoardPro 📄, | Data Mining | Statistical Analysis April 2025

- Visualizes student participation trends across **areas of interest, event types** (workshops, meetups, competitions), **modes** (online/offline) & **time periods** (monthly/quarterly/yearly).
- Provides actionable insights to improve event planning and engagement.

TickTOS ↗ , <i>RTOS System Programming OS Developement</i> A user-space simulation of a preemptive Real-Time Operating System (RTOS) with task scheduling, and real-time elevator control using priority-based scheduling.	December 2024
CHESSE GAME ENGINE ↗ , <i>ReactJS Tailwind CSS</i> - A chess engine, with the score sheet that appends the algebraic notations automatically.	July 2024 – October 2024
ATTENDIX ↗ , <i>C Programming</i> - Attendix is a terminal-based attendance management system entirely runs in C language.	March 2023 – May 2023

TECHNICAL SKILL

Programming Languages: C, C++, Python, bash, Java, asm x86, Typescript, Kotlin (basic)
Tools/Frameworks: Next.js, React.js, Node.js, Git/Github, Linux, Ollama, Cypress, Docker
Databases: MySQL, Supabase (PostgreSQL)

ROLES & RESPONSIBILITIES

Core Team Member ↗ , <i>FOSS United Coimbatore</i> - Organized Meetups in promote and build FOSS Ecosystem in Coimbatore region.	June 2025 – Present
Club Advisor ↗ , <i>FOSS-CIT</i> - Delivered a Flash task in IndiaFOSS 2025, ↗ India's Largest FOSS Conference. - Represented the Club in TOSSConf 2025 ↗ , a state level Open Source Software Conference	May 2025 – Present
Secretary ↗ , <i>FOSS-CIT</i> - Led the club by coordinating tasks and ensured fairness in teamwork - Delivered sessions on GNU/Linux, DSA, and Version Control Systems. - Team Lead & Maintainer for Open Source Projects of FOSS-CIT ↗	May 2024 – April 2025
Public Relations Manager ↗ , <i>FOSS-CIT</i> - Collaborated with participants and managed promotional activities of the Club.	May 2023 – May 2024

CO-CURRICULAR

Introduction to JavaScript Security(LFS184) ↗ , <i>The Linux Foundation</i>	November 2025
DIPLOMA in Quantum Computing ↗ , <i>QWorld Bronze 2024</i>	October 2024
WINNER in Tech Research Presentation ↗ , <i>Computing Research Day'24 - CIT</i>	September 2024

INTERESTS

Backend Development | Open Source Software Development | Operating Systems | DSA